

# SOD-123 Schottky Barrier Diodes

## MMSD301T1G, SMMSD301T1G, MMSD701T1G, SMMSD701T1G,

The MMSD301T1, and MMSD701T1 devices are spin-offs of our popular MMBD301LT1, and MMBD701LT1 SOT-23 devices. They are designed for high-efficiency UHF and VHF detector applications. Readily available to many other fast switching RF and digital applications.

### Features

- Extremely Low Minority Carrier Lifetime
- Very Low Capacitance
- Low Reverse Leakage
- AEC Qualified and PPAP Capable
- S Prefix for Automotive and Other Applications Requiring Unique Site and Control Change Requirements
- These Devices are Pb-Free, Halogen Free/BFR Free and are RoHS Compliant\*

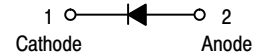
### MAXIMUM RATINGS

| Rating                                                                | Symbol    | Value       | Unit             |
|-----------------------------------------------------------------------|-----------|-------------|------------------|
| Reverse Voltage<br>MMSD301T1G, SMMSD301T1G<br>MMSD701T1G, SMMSD701T1G | $V_R$     | 30<br>70    | Vdc              |
| Forward Current (DC) Continuous                                       | $I_F$     | 200         | mA               |
| Forward Power Dissipation<br>$T_A = 25^\circ\text{C}$                 | $P_F$     | 225         | mW               |
| Junction Temperature                                                  | $T_J$     | -55 to +125 | $^\circ\text{C}$ |
| Storage Temperature Range                                             | $T_{stg}$ | -55 to +150 | $^\circ\text{C}$ |

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.



SOD-123  
CASE 425  
STYLE 1



### MARKING DIAGRAM



- xx = Specific Device Code  
 XT = MMMSD301T1G  
       SMMSD301T1G  
 XH = MMMSD701T1G  
       SMMSD701T1G  
 M = Date Code  
 ■ = Pb-Free Package

(Note: Microdot may be in either location)

### ORDERING INFORMATION

| Device      | Package              | Shipping†              |
|-------------|----------------------|------------------------|
| MMSD301T1G  | SOD-123<br>(Pb-Free) | 3,000 /<br>Tape & Reel |
| SMMSD301T1G | SOD-123<br>(Pb-Free) | 3,000 /<br>Tape & Reel |
| MMSD701T1G  | SOD-123<br>(Pb-Free) | 3,000 /<br>Tape & Reel |
| SMMSD701T1G | SOD-123<br>(Pb-Free) | 3,000 /<br>Tape & Reel |

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, [BRD8011/D](#).

\*For additional information on our Pb-Free strategy and soldering details, please download the **onsemi** Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

# MMSD301T1G, SMMSD301T1G, MMSD701T1G, SMMSD701T1G,

## ELECTRICAL CHARACTERISTICS (T<sub>A</sub> = 25°C unless otherwise noted)

| Characteristic                                                                                                                                                                              | Symbol             | Min              | Typ                         | Max                       | Unit |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|------------------|-----------------------------|---------------------------|------|
| Reverse Breakdown Voltage<br>(I <sub>R</sub> = 10 μA)<br>MMSD301T1G, SMMSD301T1G<br>MMSD701T1G, SMMSD701T1G                                                                                 | V <sub>(BR)R</sub> | 30<br>70         | –<br>–                      | –<br>–                    | V    |
| Diode Capacitance<br>(V <sub>R</sub> = 0 V, f = 1.0 MHz)<br>MMSD301T1G, SMMSD301T1G<br>MMSD701T1G, SMMSD701T1G                                                                              | C <sub>T</sub>     | –<br>–           | 0.9<br>0.5                  | 1.5<br>1.0                | pF   |
| Total Capacitance<br>(V <sub>R</sub> = 15 V, f = 1.0 MHz)<br>MMSD301T1G, SMMSD301T1G<br>(V <sub>R</sub> = 20 V, f = 1.0 MHz)<br>MMSD701T1G, SMMSD701T1G                                     | C <sub>T</sub>     | –<br>–           | 0.9<br>0.5                  | 1.5<br>1.0                | pF   |
| Reverse Leakage<br>(V <sub>R</sub> = 25 V)<br>MMSD301T1G, SMMSD301T1G<br>(V <sub>R</sub> = 35 V)<br>MMSD701T1G, SMMSD701T1G                                                                 | I <sub>R</sub>     | –<br>–           | 13<br>9.0                   | 200<br>200                | nAdc |
| Forward Voltage<br>(I <sub>F</sub> = 1.0 mAdc)<br>MMSD301T1G, SMMSD301T1G<br>(I <sub>F</sub> = 10 mA)<br>(I <sub>F</sub> = 1.0 mAdc)<br>MMSD701T1G, SMMSD701T1G<br>(I <sub>F</sub> = 10 mA) | V <sub>F</sub>     | –<br>–<br>–<br>– | 0.38<br>0.52<br>0.42<br>0.7 | 0.45<br>0.6<br>0.5<br>1.0 | Vdc  |

TYPICAL CHARACTERISTICS  
MMSD301T1G, SMMSD301T1G

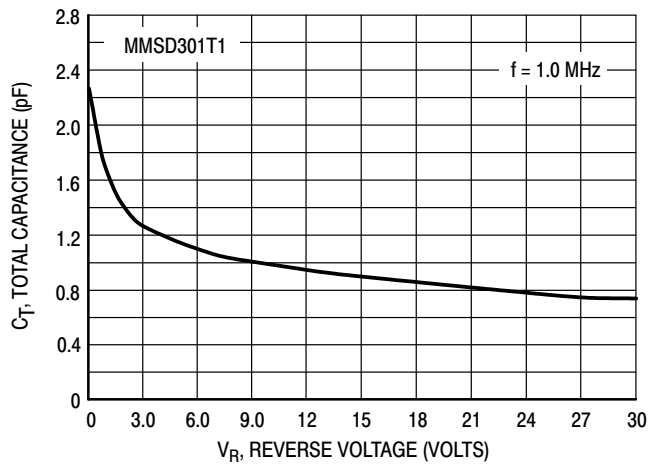


Figure 1. Total Capacitance

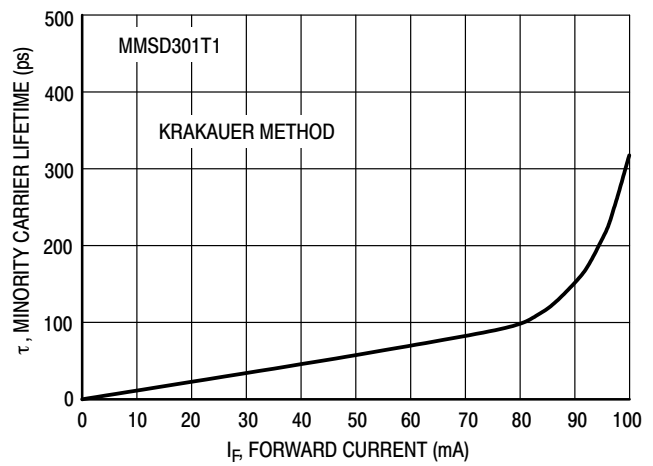


Figure 2. Minority Carrier Lifetime

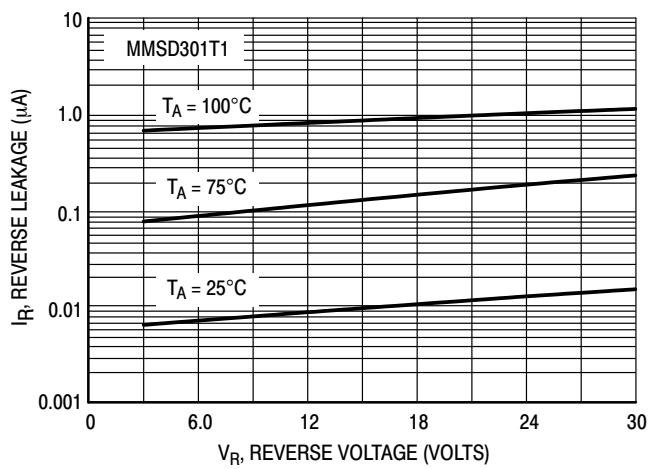


Figure 3. Reverse Leakage

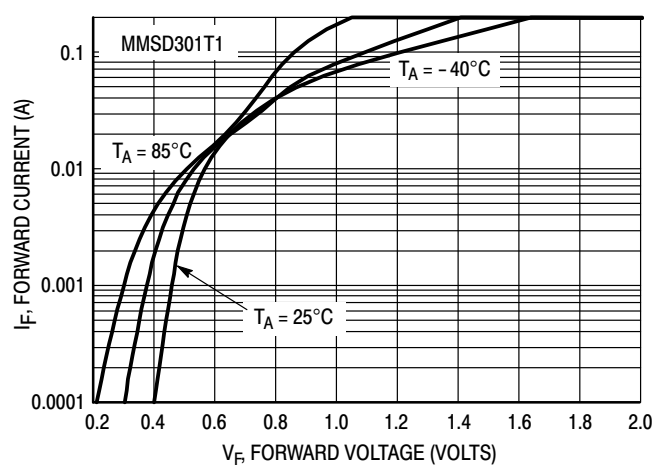


Figure 4. Forward Voltage

TYPICAL CHARACTERISTICS  
MMSD701T1G, SMMSD701T1G

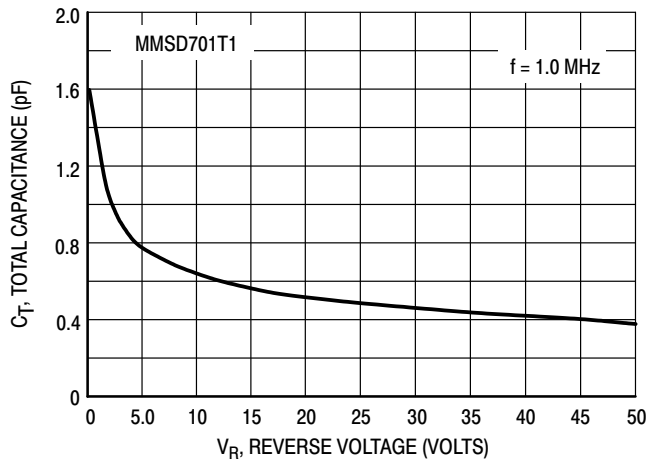


Figure 5. Total Capacitance

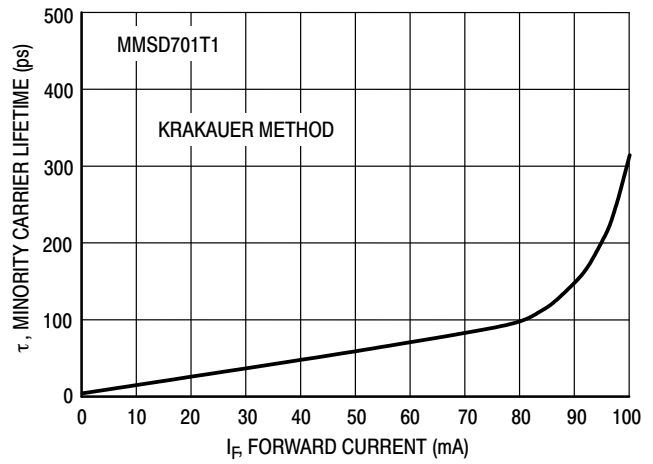


Figure 6. Minority Carrier Lifetime

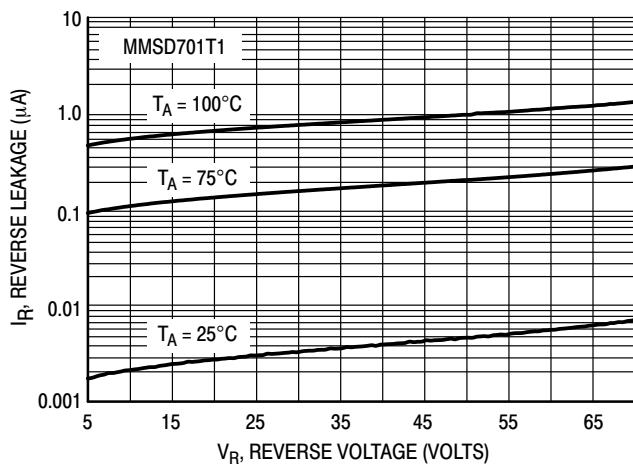


Figure 7. Reverse Leakage

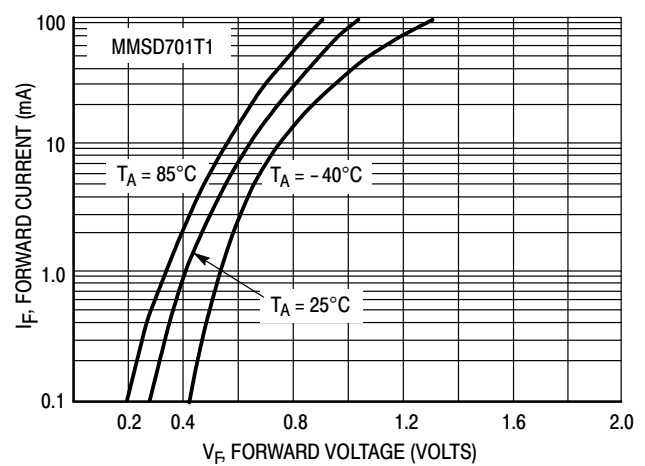
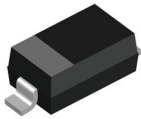
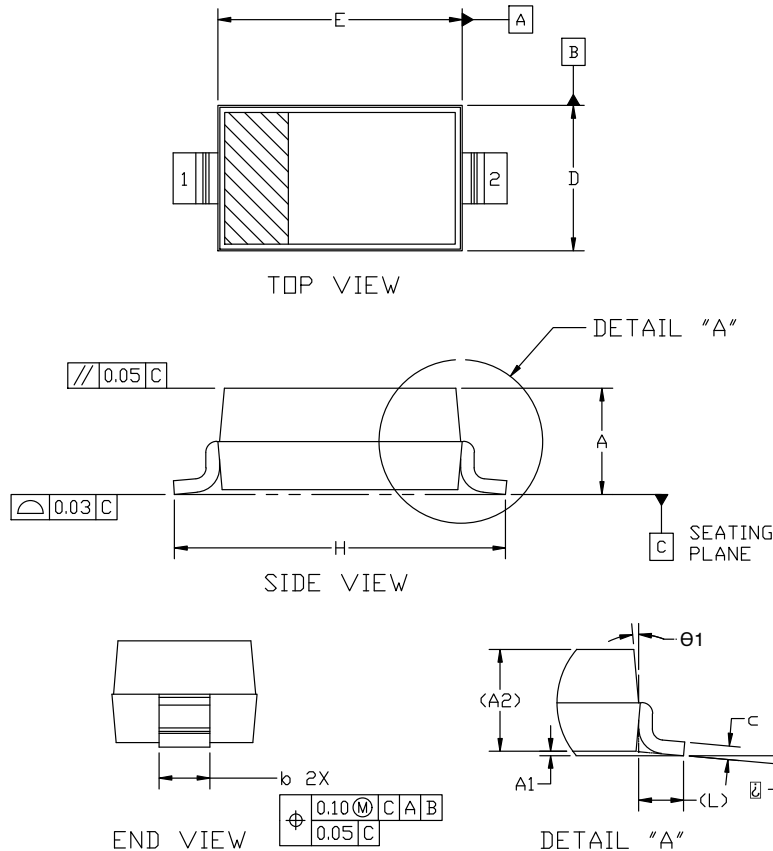


Figure 8. Forward Voltage


**SOD-123 2-LEAD, 1.60x2.69x1.16**  
**CASE 425**  
**ISSUE H**

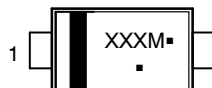
DATE 29 FEB 2024



## NOTES:

1. DIMENSION AND TOLERANCING PER ASME Y14.5M, 2018
2. CONTROLLING DIMENSION: MILLIMETERS

| DIM | MILLIMETER |      |      |
|-----|------------|------|------|
|     | MIN.       | NOM. | MAX. |
| A   | 0.94       | 1.17 | 1.35 |
| A1  | 0.00       | 0.05 | 0.10 |
| A2  | 1.16 REF.  |      |      |
| b   | 0.51       | 0.61 | 0.71 |
| c   | —          | —    | 0.15 |
| D   | 1.40       | 1.60 | 1.80 |
| E   | 2.54       | 2.69 | 2.84 |
| H   | 3.56       | 3.68 | 3.86 |
| L   | 0.25 REF.  |      |      |
| θ   | 0°         |      | 10°  |
| θ1  | 0°         |      | 10°  |

**GENERIC MARKING DIAGRAM\***


XXX = Specific Device Code  
M = Date Code  
▪ = Pb-Free Package

(Note: Microdot may be in either location)

\*This information is generic. Please refer to device data sheet for actual part marking. Pb-Free indicator, "G" or microdot "▪", may or may not be present. Some products may not follow the Generic Marking.

STYLE 1:  
PIN 1. CATHODE  
2. ANODE

**RECOMMENDED MOUNTING FOOTPRINT**

\*For additional information on or Pb-Free strategy and soldering details, please download the ON Semiconductor Soldering and Mounting Techniques Reference manual SOLDERM/D.

|                         |                                       |                                                                                                                                                                                  |
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| <b>DESCRIPTION:</b>     | <b>SOD-123 2-LEAD, 1.60x2.69x1.16</b> | <b>PAGE 1 OF 1</b>                                                                                                                                                               |

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